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Handover management in beyond 5G HetNet topologies with unbalanced user distribution ☆

Abdussamet Hatipoglu [a], Mehmet Akif Yazici [a], Mehmet Basaran [b,a,*], Mine Ardanuc [a], Lutfiye Durak-Ata [a]

[a] Information and Communications Research Group (ICRG), Informatics Institute, Istanbul Technical University, Istanbul 34467, Turkey
[b] 6GEN LAB, Next-Generation R&D, Network Technologies, Turkcell, Istanbul 34854, Turkey

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ABSTRACT

The increase in user mobility and density in modern cellular networks increases the risk of overloading certain base stations in popular locations such as shopping malls or stadiums, which can result in connection loss for some users. To combat this, the traffic load of base stations should be kept as balanced as possible. In this paper, we propose an efficient load balancing-aware handover algorithm for highly dynamic beyond 5G heterogeneous networks by assigning mobile users to base stations with lighter loads when a handover is performed. The proposed algorithm is evaluated in a scenario with users having different levels of mobility, such as pedestrians and vehicles, and is shown to outperform the conventional handover mechanism, as well as another algorithm from the literature. As a secondary benefit, the overall energy consumption in the network is shown to be reduced with the proposed algorithm.

1. Introduction

In recent years, Heterogeneous Networks (HetNets) consisting of different types of Base Stations (BSs) including macro BSs (mBSs) and small BSs (sBSs) have played an important role in improving network performance through densification in certain locations. The deployment of a large number of BSs in Beyond 5G (B5G) networks is expected to improve the overall system performance by both extending coverage and providing a better user experience [1]. Mobility management is also crucial for the novel services envisioned in B5G networks to maintain a seamless user experience in a large variety of applications. To cope with the dramatic growth of mobile traffic, B5G networks are envisioned to provide a higher Quality of Service (QoS) [2]. Densifying the network by increasing the number of BSs reduces the load per BS (which can be defined either in terms of traffic demand or the number of connected users) and thus, increases the QoS. Due to the low cost and easy deployment of sBSs, they are highly preferred in HetNets [3].

When a mobile User Equipment (UE) leaves the coverage area of the serving BS, the UE must be handed over to a neighboring cell without disrupting the connection [4]. However, the high mobility of users in ultra-dense HetNets leads to an increase in the number of Handovers (HOs) for mobile users. The uneven nature of the user distribution in the network due to certain attraction points in a location can lead to an unbalanced load on the BSs. As the number of HOs increases, Load Balancing (LB) is inevitably required to avoid overloading at specific BSs. Typically, when the traffic load of a mBS increases dramatically, the HO management algorithm directs the traffic load to the BSs with lighter loads. Therefore, HO and LB are always important operations to ensure a fair QoS for UEs. In addition, HO and LB are seriously affected by connection problems such as HO Failures (HOFs) in the network [5]. Although some HO parameters such as Time-To-Trigger (TTT) and HO Margin (HOM) can be modified to reduce HOFs, changing the configurations too frequently can negatively affect HO performance.

In this paper, an LB-aware HO algorithm is proposed for B5G HetNets in order to improve the network performance in terms of load distribution, as well as HO success and the number of HOs. The proposed algorithm is evaluated in a stochastic geometry-based HetNet topology, which is closer to the real BS deployment compared to the regular hexagonal cell tessellations. User mobility behavior is also taken into account by classifying cells as ordinary or attractive, based on the probability that a user chooses to move into each cell, as well as the dwell time of the user in each cell.

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* Corresponding author: 6GEN LAB, Next-Generation R&D, Network Technologies, Turkcell, Istanbul 34854, Turkey.
E-mail addresses: hatipoglua@itu.edu.tr (A. Hatipoglu), yazicima@itu.edu.tr (M.A. Yazici), mehmet.basaran@turkcell.com.tr, mehmetbasaran@itu.edu.tr (M. Basaran), ardanuc@itu.edu.tr (M. Ardanuc), durakata@itu.edu.tr (L. Durak-Ata).

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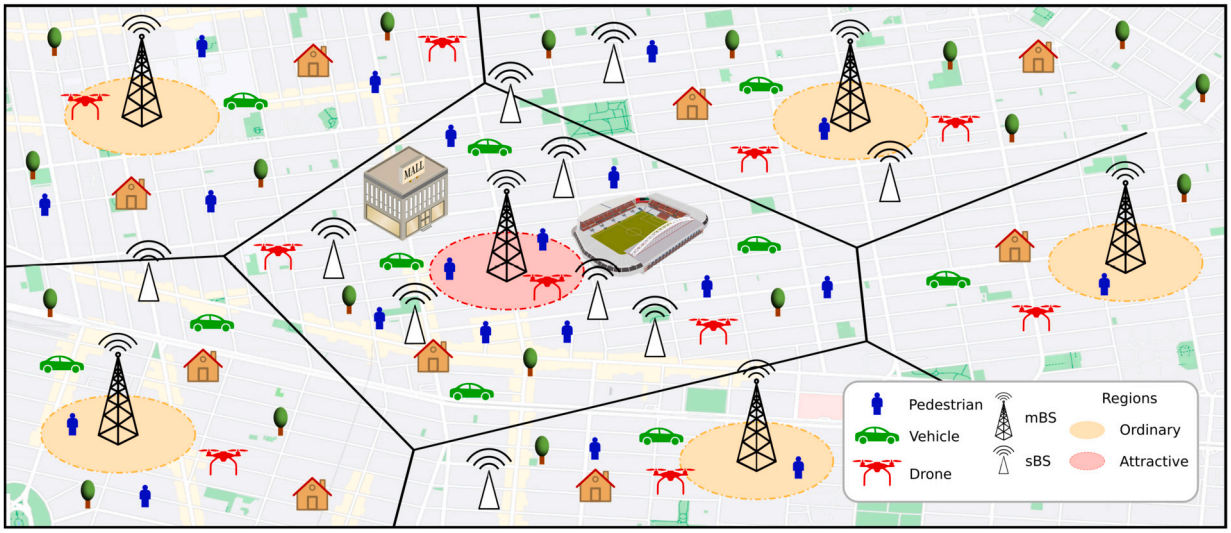

Fig. 1. An illustration of a sample HetNet structure consisting of mBSs, sBSs, and UEs. There are certain places such as stadiums and malls where people gather more densely as in real life. In addition, different user types such as pedestrians, vehicles, and drones can be observed. In some cases, overloading of BSs can occur in the center of a city or in extreme conditions such as derby matches and huge concerts.

1.1. Related work

Recently, many studies have focused on HO and related issues in HetNets, and introduced algorithms to improve network performance [6–18]. An algorithm to mitigate frequent HOs for fast-moving users in HetNets is proposed in [6]. This algorithm adjusts the HO Control Parameters (HOCP) based on the ratio of HO ping-pong, where UE is handed over to the mBSs in case of high mobility. In addition to [6], a dynamic HOCP algorithm is proposed in [7] to investigate HO types that cause HOF. This algorithm also adjusts TTT in addition to HOM. The results show that the dynamic HOCP algorithm achieves lower HOF and HO ping-pong rates compared to the fixed HOCP. Furthermore, in [8,9], the authors present an optimization problem with a solution that guarantees proportional fairness and significantly reduces the handover rate. They propose a deep reinforcement learning algorithm to solve the problem, which collectively considers the decisions of when to hand over and which BS to assign to a user.

A cluster-based mobility load-balancing algorithm for heterogeneous LTE networks is proposed in [10]. The algorithm dynamically creates cell clusters and performs load balancing on these clusters, taking into account the overloaded cells and their neighbors. UE association algorithms and traffic offloading are combined by the authors in [11], using selected sBSs where the working region is divided into center and edge regions in order to achieve a more stable network.

Markov chains are one of the tools used to model and analyze user mobility and the HO process. In [12], a non-homogeneous discrete-time Markov chain is used to model the relative position and the evolution of the HO process of a mobile user in a scenario with a single mBS and a femto-BS. Cell loads and user trajectory are considered in the HO decision. However, it is difficult to generalize the model to larger networks (with more UEs and BSs) because the state space of the Markov chain would grow exponentially with the number of femto BSs. In [13], vertical HOs are studied using a two-dimensional Markov chain approximation in a scenario where WiFi connections are used as the main path, while multi-path TCP is used to maintain a cellular connection as a secondary failover path. The signal qualities of both connections are monitored and define the states of the two-dimensional Markov chain. The main goal of [13] is energy efficiency, in contrast to LB.

In [14], the authors study the load balancing problem in a cellular network and propose an algorithm to resolve the mismatch between the distribution of network resources and the traffic demand by transferring the UEs of an overloaded cell to a neighboring cell. A service-based mobility LB algorithm, which is considered as an operator utility for HO operation in 5G networks is proposed in [15]. In this algorithm, an LB efficiency factor is introduced to take into account the loads of the neighboring cells and the edge UEs of the overloaded cell. In [16], the authors propose a mobility LB algorithm that adjusts the capacity of BSs depending on the loads of neighboring cells by considering the network load state and using load estimation. UEs at cell boundaries are identified and handed over to the adjacent cell.

Recently, in [17], we proposed an algorithm that uses user speed and Received Signal Reference Power (RSRP) to adjust HOM and TTT in a hexagonal cellular topology. The proposed algorithm balances the load in terms of number of connected UEs among neighboring BSs. The simulation results show that the rates of HO ping-pong and HOF are significantly reduced by the proposed algorithm compared to the conventional handover mechanism. In the literature, stochastic geometry-based HetNet topology models are proposed to mimic the real BS deployment [18].

1.2. Main contributions

In this study, we propose an efficient LB-aware HO algorithm for highly dynamic B5G networks. The main goal of this study is to propose a HO mechanism which employs LB to a certain extent in order to improve the HOF performance as well as the power consumption of the network in a much more realistic setting in terms of BS placement and user mobility behavior. The contributions of this study can be summarized as follows.

i The proposed handover mechanism considers the loads of the candidate BSs and aims to balance the loads of the BSs, in terms of the number of users connected, across the network.

ii The proposed handover mechanism is evaluated in a generally more realistic setting (illustrated in Fig. 1) compared to the studies in the literature. The components of the evaluation scenario are as follows:
- Instead of a regular hexagonal cell tessellation, we assume a relatively more realistic network topology by employing Poisson-Voronoi tessellation. The BSs are randomly distributed according to a Poisson Point Process (PPP) [19]. In addition, BSs that fall within the exclusion regions of mBSs in a Poisson Hole Process (PHP) model [20] are removed to account for interference avoidance in a real network infrastructure.

- Each macro cell is classified as ordinary or attractive, based on the likelihood that a user will choose to move into a cell, as well as the amount of time the user spends in that cell. This reflects the real behavior of users, who tend to congregate around certain locations such as shopping malls, business or cultural centers, sports arenas, or designated safe places in case of emergencies such as fires or natural disasters, potentially leading to the overloading of BSs in these attractive regions.
- Users are assumed to belong to one of the three categories: pedestrians, vehicles, and drones, which are mainly classified according to their speed.

iii The proposed handover mechanism is shown to improve performance in terms of the number of handovers, handover success, and load balance across BSs.

iv It is shown that the overall energy consumption in the network is reduced with the proposed algorithm.

The rest of the paper is organized as follows. In Section 2, the system model is described in terms of the spatial distribution of base stations and users, the mobility model, and the power consumption model. The proposed algorithm is explained in Section 3. The simulation parameters and environment, the numerical results obtained, and their interpretations are presented in Section 4. The paper is concluded in Section 5.

2. System model

Multi-tiered networks consisting of mBS and sBSs with different user profiles, both in terms of mobility and data traffic characteristics. In this study, we consider enhanced mobile broadband access, where users expect high data rates with significant traffic volumes. The spatial distribution of the base stations is modeled using stochastic point processes instead of a fixed hexagonal grid. The mobility patterns of the UEs depend on both their intrinsic properties, such as pedestrian, vehicular, or even drone, and the environment. In such a system, users with different degrees of mobility, as well as connected devices such as driverless cars and drones, can be served. At a given location for a given period of time, there may be hot zones that attract users more than other locations. This could be due to the presence of places such as shopping malls, concert or sports venues and stadiums, and historical attractions. User behavior may also be time-dependent, such as business districts being more popular on the weekdays during business hours, or users congregating in certain safe locations during fires or natural disasters. A sample scenario is shown in Fig. 1.

2.1. Spatial distribution of the base stations

Instead of the uniform hexagonal tessellation, we consider a more realistic distribution of BSs on the two-dimensional plane. In [21], the authors propose a multi-tier cellular network in which point processes, such as three different Mat'ern Hard-core Process (MHCP) variants and the PHP, are used to capture intra- and inter-tier dependence in the BS locations. In our model, mBS locations are modeled using a type II MHCP, while sBS locations are modeled using a PHP [22,23]. The reason for choosing an MHCP to model the distribution of the mBSs is that the mBSs are deployed so that they are not too close to each other. This reduces interference and allows for more efficient frequency reuse. In addition, a PHP is preferred to model the distribution of the sBSs so that they are not placed in extreme proximity to the mBSs [24].

The PPP Ψ is a process where the number of points contained in a region B has a Poisson distribution with intensity $\Lambda(B)$. If $\Lambda(B) = \lambda\,|B|$, where $|B|$ is the area of B, the process is called a homogeneous PPP.

Let Ψ_0 and Ψ_1 be two independent PPPs with intensities λ_0 and λ_1, respectively. Also let Ξ_D be the union of disks of radius D centered at the points belonging to Ψ_1:

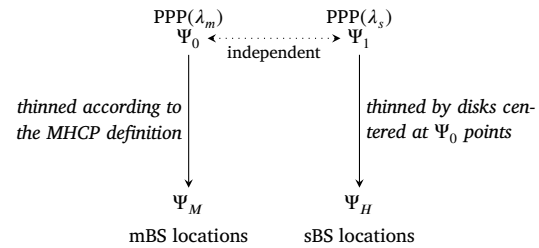

Fig. 2. The process of producing mBS and sBS locations.

$$\Xi_D = \bigcup_{y\in\Psi_1} \{x : ||x-y|| \le D\}$$

The process $\Psi_H = \Psi_0 \backslash \Xi_D$, obtained by thinning Ψ_0 by censoring the points falling inside Ξ_D is called the PHP. In this context, we call Ψ_1 the *generating PPP*, and Ψ_0 the *thinning PPP*.

Consider again the PPP Ψ_0, with intensity λ_0. Each point $x \in \Psi_0$ is assigned a mark $m(x)$, which is a random value uniformly distributed on $[0, 1]$, independent of all other mark values. Then, all points a neighbor with a smaller flag within a distance of R are marked. The MHCP type II [25, pp. 46–49], Ψ_M, is obtained by simultaneously removing the flagged points from Ψ_0:

$$\Psi_M = \Psi_0 \backslash \bigcup_{y\in\Psi_0} \{x : ||x-y|| \le R,\ m(x) < m(y)\}$$

In this paper, we consider a HetNet with two layers consisting of mBSs and sBSs. The locations of the mBSs are assumed to be distributed according to a MHCP Ψ_M, with an underlying PPP Ψ_0 with an intensity of λ_m. On the other hand, the locations of the sBSs are assumed to be distributed according to a PHP, where the generating PPP has an intensity of λ_s, and the thinning PPP is the same Ψ_0 as the underlying Ψ_M. In other words, first, a PPP Ψ_0 with an intensity of λ_m is generated. Then, Ψ_0 is thinned by removing some of the neighboring points as described in the definition of the MHCP, to produce the mBS locations. Next, a second PPP with an intensity of λ_s, Ψ_1, independent of Ψ_0 is generated, and then thinned by removing the points within a distance of D from the points in Ψ_0. This process is summarized in Fig. 2.

Due to the thinning process, the effective node densities of Ψ_0 and Ψ_1 are less than λ_m and λ_s, respectively. In this case, the effective density of the mBSs is [26]

$$\lambda_m^{(e)} = \frac{1-\exp(-\lambda_m \pi R^2)}{\pi R^2} \tag{1}$$

whereas the effective density of the sBSs is

$$\lambda_s^{(e)} = \lambda_s\ \exp(-\lambda_m \pi D^2) \tag{2}$$

2.2. Spatial distribution of the users

To achieve a more realistic model, the number of UEs in the system is chosen to be proportional to the population density of the site. For each region, mBSs have a radius R_a around them, which represents an attractive radius, creating an attractive zone where people gather when a particular activity takes place or an extreme situation occurs. R_a is different for each region and is between $R_a^{\min}$ and $R_a^{\max}$. UEs initially select a region based on the density multiplier vector, $\vec{\rho}$. Each element of $\vec{\rho}$, denoted ρ_i, is calculated by multiplying $R_{a,i}$ and λ_u of the i-th region, as given below:

$$\rho_i = \begin{cases} m\,\lambda_u\,R_{a,i}^2, & i = 1 \\ \lambda_u\,R_{a,i}^2, & i \ne 1 \end{cases} \tag{3}$$

Accordingly, λ_u in the central region is m times larger than that of the other regions. The central region is denoted with $i = 1$, and the other regions $i \ne 1$. As the central region is more densely populated, the number

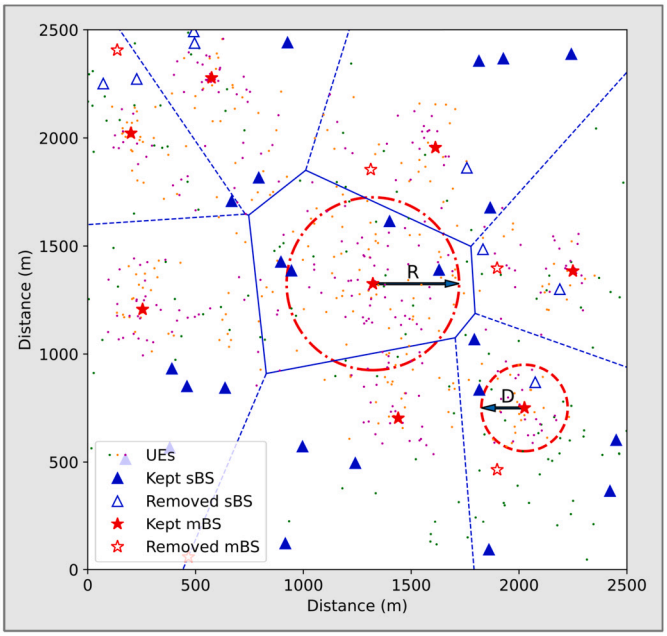

Fig. 3. An illustration of designed HetNet topology (stars, triangles, and dots represent mBSs, sBSs, and mobile users, respectively).

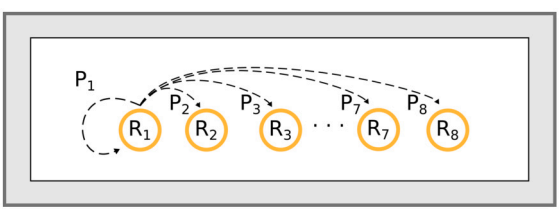

Fig. 4. Markov chain for the transition probabilities (R_i and P_i represent the region and the probability of moving to that region, respectively).

of UEs in this region is also higher than in other regions. Furthermore, different types of user models have been defined, namely pedestrian, vehicle and drone, which have low, medium and high speeds respectively. Afterwards, UEs are then randomly distributed across the selected regions. The total number of UEs in the system is denoted with N_{UE}.

A sample scenario of an ultra-dense HetNet with two layers, consisting of mBS and sBS is shown in Fig. 3. Users with different speeds are shown in different colors. Empty triangles and stars denote removed BSs, while filled ones denote kept BSs.

2.3. Mobility model

All regions have seven neighbors with stable modes according to the scenario explained in the previous subsections and illustrated in Fig. 3. UEs in each region have eight (seven neighboring regions and one current region) different transition probabilities. These region transitions can be modeled using a Markov chain as shown in Fig. 4. State transitions represent the physical movement of the UE to the corresponding neighboring region with probability P_i. Physical transfers of UEs are then are performed once the target region is selected according to the transition probabilities.

In this scenario, the UEs are stationary at a given point for a random amount of time with an exponential distribution with mean τ. At the end of this time, they select a new target and move to the new target point randomly chosen within the selected region at a speed of V.

The transition probabilities of the mobile UEs can be represented by the transition matrix, **T**. Since these probabilities represent the selection probabilities at the end of the waiting times, the resulting Markov chain is a discrete-time Markov chain. The components of the matrix **T** can be expressed as

$$\mathbf{T}_{ij} = \mathbb{P}\left(S_i \to S_j\right) = \mathbb{P}\left(X_{t+1} = S_j \mid X_t = S_i\right) \tag{4}$$

which is the transition probability to move from state S_i at time index t to state S_j at time index $(t+1)$. In this context, $\mathbf{T}_{ij}$ indicates the probability of choosing region j as a target for the UE currently residing in region i [27].

The diagonal elements of **T** are denoted by $k_i \in (0,1)$. Since the central region (enumerated to 1 without loss of generality) is more attractive, k_1 is chosen higher compared to other regions, as given by Equation (5). Hence, k_i is expressed conditionally as k_c for the central region and k_o for other regions, i.e.,

$$k_i = \begin{cases} k_c, & i = 1 \\ k_o, & i \neq 1 \end{cases} \tag{5}$$

where $k_c > k_o$. The elements of **T** are then calculated as

$$\mathbf{T}_{ij} = \begin{cases} k_i, & i = j \\ \dfrac{1-k_i}{N-1}, & i \neq j \end{cases} \tag{6}$$

where N represents the number of regions.

Using the steady-state solution of the Markov chain, denoted $\vec{r}$, the long run probability of finding a UE in any region can be found. Then, the expected number of UEs in region i can be expressed as

$$E[N_{UE,i}] = r_i\, N_{UE} \tag{7}$$

Accordingly, mBS capacity thresholds are assigned for LB using a threshold factor β as

$$\delta_{th}^{i} = E[N_{UE,i}]\,\beta_i \tag{8}$$

2.4. Power consumption model

Base station power consumption is an important aspect of cellular network design and deployment and has been extensively studied [28–32]. The base station power consumption model consists of a static and a dynamic power component. The static component, the power consumption of a BS in the idle mode, is the minimum energy required for the normal operation of the BS. In contrast, the dynamic power component depends on the traffic load. Most studies propose the following linear model:

$$P_j = \Delta_j\, P_j^T(\rho_j(t)) + P_j^s \tag{9}$$

where Δ_j is the slope of load-dependent power consumption, P_j^T is the transmission power depending on the traffic load $\rho_j(t)$, and P_j^s is the static power component. According to Equation (9), the power consumption for BS_j is a function of traffic load. The power model can be adjusted for any kind of BS (macro or small) by modifying the model parameters.

3. The proposed algorithm

The algorithm we propose is based on the RSRP levels of the UEs and the instantaneous loads of the BSs. The Path Loss (PL) is considered when calculating the RSRP. The mathematical equations for PL are given in [33–35]. Finally, the RSRP is calculated as

$$\text{RSRP (dBm)} = P^T + P^G - \text{PL} \tag{10}$$

where P^T is the transmitter power and P^G is the receiver antenna gain. As P^T is the transmitter power and we assume that it is chosen compatible with the standards, it is reasonable to assume that the UE is aware of its value. In any case, even if the transmitter uses a non-standard value, this value can be assumed to be communicated over the control

Algorithm 1: LB-Aware HO Algorithm.

```
Inputs: λ, ρ⃗, k, N, and β
Obtain T, according to Eq. (5) and (6)
for each mBS as mBS_i do
    δ^i_th = λ_i * β
    Set mBS_i threshold to δ^i_th
end
for each UE do
    Get the BSs according to RSRP in decreasing order for i in [1,2,3]
      do
        if BS_i state is available then
            BS_i connects UE to itself, and exit
    end
    for i in [1,2,3] do
        if BS_i state is semi-available then
            BS_i connects UE to itself, and exit
    end
    for each BS do
        The first available or semi-available BS connects UE to itself
    end
end
```

Table 1
The availability states of the BSs (Load is taken as the number of connected UEs).

Status	Available	Semi-Available	Unavailable
Load of BS	$< \delta^i_{th}$	δ^i_{th} up to Full	Full

channel in pilot/beacon messages. As P^G is the receiver antenna gain, it is inherently known by the UE.

To perform HO procedures, RSRP values must be measured by UEs. Each UE periodically measures the RSRP values of neighboring cells. It sends measurement reports to the BS only when the RSRP of a neighbor cell has a better signal level than the signal of the serving cell. Based on these measurement reports, the serving BS makes a decision on whether to perform HO. If the RSRP value of one of the neighboring cells is greater than the RSRP of the serving cell by HOM for a duration of TTT, the BS sends the HO command to the UE. The condition that triggers HO is expressed as

$$RSRP_{neigh} > RSRP_{serv} + \text{HOM} \tag{11}$$

where P_{serv} indicates the RSRP value of the connected cell, while P_{neigh} represents the RSRP value of the neighbor cell with the strongest signal. If the condition given in (11) continues to be satisfied for a duration of TTT, then HO is performed.

In the proposed algorithm, described in Algorithm 1, the capacity threshold value, δ^i_{th}, is obtained for mBS$_i$ by multiplying the expected number of UEs per region and the parameter β. For each mBS, there are three states of availability, namely, *available*, *semi-available*, and *unavailable*. There are also available and unavailable states for the sBSs. In the available state, a BS has fewer UEs connected to itself than the threshold δ_{th}. In the semi-available state, there are more UEs connected to the BS than δ_{th}, but fewer UEs than its total capacity. Otherwise, the BS is in the unavailable state, i.e. it has reached its full capacity. The summary of the availability states of the BSs is given in Table 1.

When a UE needs to perform HO, it tries to connect to the best candidate in terms of RSRP if this BS is in the available state. If not, the UE tries the next two best candidates in order, connecting to the first one that is in the available state. If none of the three BSs are in the available state, it tries to find a BS among the best three in the semi-available state. If this is also not possible, the remaining BSs are searched in the order of descending RSRP values for the first available or semi-available BS.

Table 2
Simulation parameters.

	Parameters	Value		
Global Parameters	Simulation repetitions	10000		
	Simulation time (s)	1800		
	Time-to-triger (ms)	100		
	Simulation area (m^2)	2500×2500		
	Threshold factor, β	$\{0.5, 0.6, \ldots, 1\}$		
		mBS	**sBS**	
BS Related Parameters	Density, λ_m, λ_s (1/m^2)	2×10^{-6}	5×10^{-6}	
	Transmission power, P^T (dBm)	43	21	
	Slope of load-dependent power consumption, Δ	4.7	4.0	
	Static power component, P^s (Watt)	130	6.8	
	Capacity of BS (number of users)	200	50	
	Exclusion zone radius, R, D (m)	500	200	
	Antenna gain P^G (dBm)	10	10	
		Center	**Others**	
Region Related Parameters	User density, λ_u (1/m^2)	2×10^{-3}	1×10^{-3}	
	Attraction factor, m	2	-	
	Probability of staying in the same region (k_c, k_o)	1/2	1/3	
	Attractive radius lower bound, $R^{\min}_a$ (m)	300	150	
	Attractive radius upper bound, $R^{\max}_a$ (m)	300	300	
		Pedestrian	**Vehicle**	**Drone**
User Type Related Parameters	UE speed, V (m/s)	1	10	25
	Average dwell time, τ (s)	60	30	10

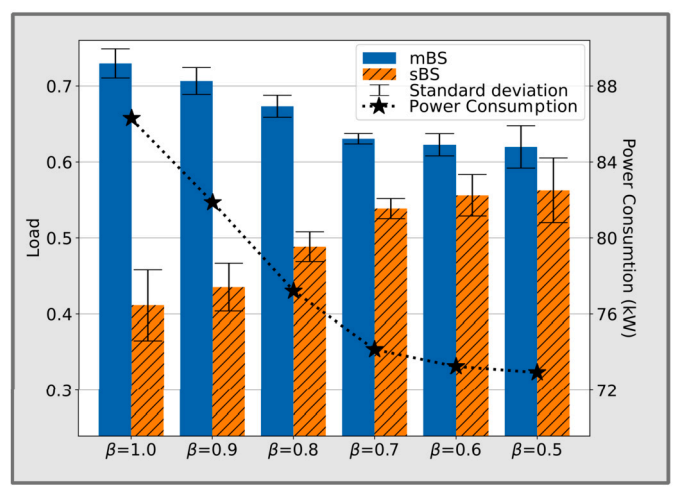

Fig. 5. Load distribution of the network and power consumption according to different β values.

4. Performance evaluation

In this section, we present simulation results to demonstrate the performance of the proposed algorithm. For this purpose, a standalone simulator is developed in Python. The simulation parameters are listed in Table 2 including the parameters related to BS, region and user type. As a benchmark, we compare the performance of the proposed algorithm with the conventional HO mechanism, as well as with the adaptive LB algorithm proposed in [16], which we also implement under the same scenario.

We first examine the effect of β, which is varied between 0.5 and 1, on the load distribution across the BSs. The average load distributions of mBSs and sBSs (along with their standard deviations), as well as the average power consumption values, at the end of 10000 runs, are shown in Fig. 5. The gap between the loads of mBSs and sBSs ob-

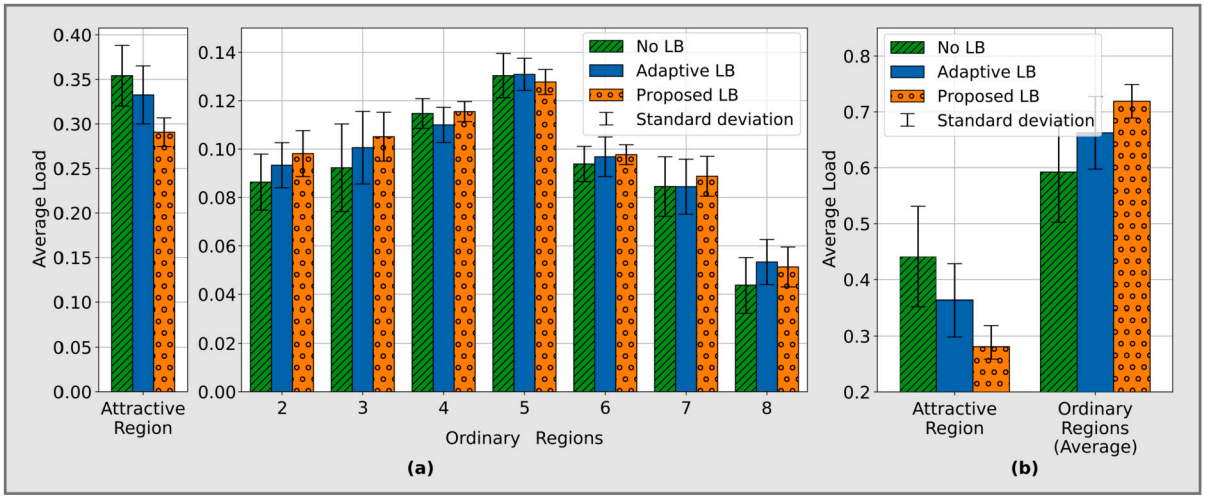

Fig. 6. The comparison of the proposed LB-Aware HO Algorithm with other methods: a) The load of the BSs in each region, taken from one sample simulation. (The load in the center is shifted to other regions by the proposed algorithm.) b) The average load of BSs in the center and other regions, as a result of 10000 simulation runs.

served in the network is reduced by decreasing β, which shows that sBSs are more heavily loaded compared to mBSs. This means that the variance of the load among the BSs in the network is reduced, thus, the load distribution among the BSs in the network has become more balanced.

Furthermore, Fig. 5 also shows the performance of the proposed algorithm in terms of load variance of BSs in the network. For example, when β is selected as 0.7, the standard deviation becomes the smallest. Although the gap becomes smaller, the variance becomes higher when β is chosen less than 0.7. Furthermore, as shown in Fig. 5, the total amount of energy consumed in the system decreases as the use of SBS increases. Thus, according to the transmission power values of the BSs in Table II, the increasing use of sBSs, which is inversely proportional to β, reduces the total power consumption. The end of the sharp decrease in power consumption at $\beta = 0.7$ makes this value the first candidate as the best β, which can be selected as the near-optimal simulation parameter value.

In order to compare the proposed algorithm with the conventional and the adaptive HO mechanisms, we assume that all users behave similarly in terms of their average traffic demand and average channel occupancy duration, which means that the energy consumption is proportional to the number of users per each BS, either macro or small. Although it would be possible to assume different user classes in this respect, this would add very little insight and is therefore not pursued in this work. For evaluating the overall performance of the proposed algorithm, we compare it with the conventional and the adaptive LB algorithms by considering the load distribution. To simplify the figures, we denote the proposed LB-aware HO algorithm as “Proposed LB”, the adaptive load balancing algorithm as “Adaptive LB”, and the conventional algorithm as “No LB” since it does not use any LB methods.

The load distribution of the central region and seven neighboring regions, as explained in Section 2, obtained after 10000 simulations are shown in Figs. 6a and 6b, respectively. It can be observed that the load of the BSs in the central region is transferred to the other regions by the proposed algorithm. The reason for this transfer is that the mBSs in the center generally operate in the semi-available state, and the UEs trying to perform HO are instead redirected to the available sBSs in their vicinity. The adaptive LB algorithm also achieves this to a certain extent, but is not as successful as the proposed algorithm. Fig. 6a shows that the loads on all the BSs are much more balanced (close to each other) when the proposed algorithm is used, compared to the other two methods where we observe larger differences in loads between regions and, consequently, a higher load on the mBS in the attractive region. It should be noted that the proposed method does not reduce the total overall load (which is a system parameter), but the differences between the loads among different BSs. Fig. 6a shows the results of a single simulation as an example, whereas Fig. 6b shows the average over 10000 simulations, which shows that the proposed model leads to a significant improvement in the availability of the mBS located at the attractive region by keeping its load low (0.285 with a standard deviation of 0.03, as opposed to 0.373 with a standard deviation of 0.065 for Adaptive LB, and 0.438 with a standard deviation of 0.09 for the standard (No LB) operation).

Next, we investigate the effect of the total system load on the performance. Fig. 7a shows the total load percentage of mBSs and sBSs for the three different algorithms for different densities. The gap between the load of the mBSs and the sBSs for the proposed algorithm is less than the other two methods. The HOF observed when using each method for different UE densities is shown in Fig. 7b. As the density increases, increases and therefore the number of connection failures increases. For this reason, the HOF values are higher for high density scenarios. It can be seen that the proposed LB algorithm has better HOF values than the other approaches. As the proposed algorithm provides better load distribution, HOFs occur less frequently as UEs trying to perform HO manage to find BSs that are available in their vicinity.

Fig. 8a shows the number of HOs of a user moving at the boundaries of the central region. According to the simulation results performed at different speeds, the proposed algorithm performs HO on average 66 times, while the Adaptive LB and NO LB algorithms perform HO on average 123 and 129 times, respectively. Accordingly, the proposed algorithm, which is designed to balance the load in HetNets, also outperforms other approaches in terms of the number of HOs performed. Since a user consumes more energy while performing HOs, the proposed algorithm also saves energy. Fig. 8b shows the HOF values are given for types of users. As the speed of drones is higher than other user types, the HOF value is higher. The speed of pedestrians is low and therefore the HOF values are also low. Furthermore, the proposed algorithm has significantly lower HOF values for all user types.

The proposed algorithm is also stress-tested in a scenario where the system load suddenly increases, such as after an earthquake or during a penalty kick at a football match. The load change of the mBSs for the three methods is shown in Fig. 9, in response to a sudden 10% increase in the number of UEs at time $t = 10$. The proposed algorithm seems to handle the sudden increase in the load much better than the other methods.

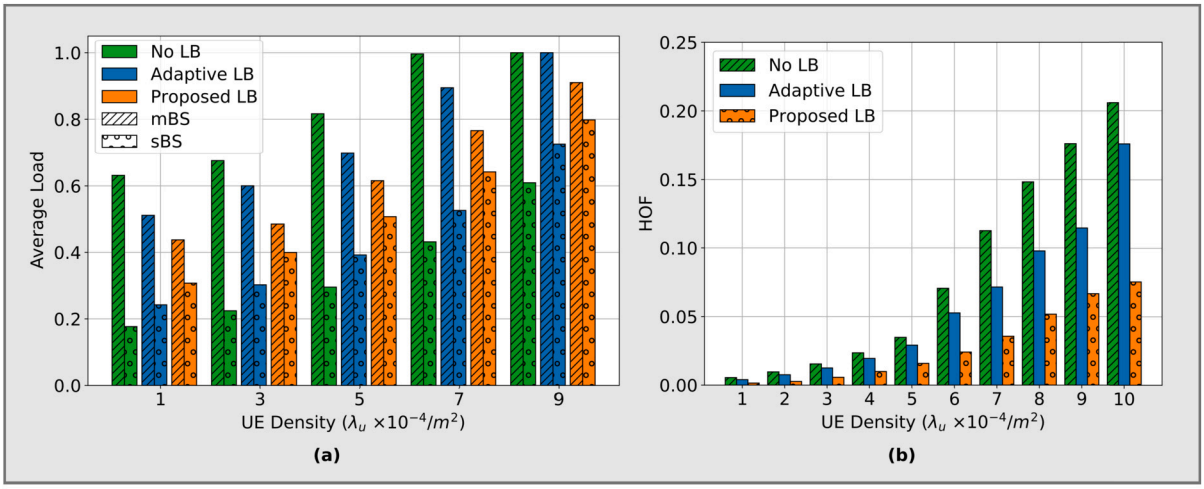

Fig. 7. The comparison of the proposed LB-Aware HO Algorithm with other approaches, obtained as a result of 10000 simulations: a) Load distribution of the BSs according to different user densities. b) HOF for each method for user different densities.

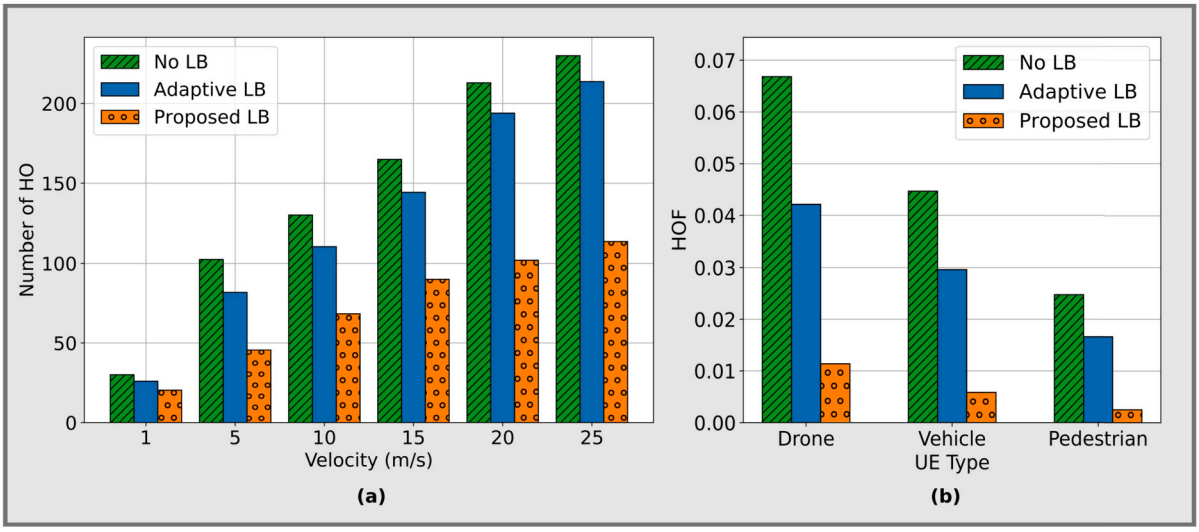

Fig. 8. a) Number of HOs for a user moving at the boundaries of the central region with varying speeds. b) HOF figures with respect to different UE types.

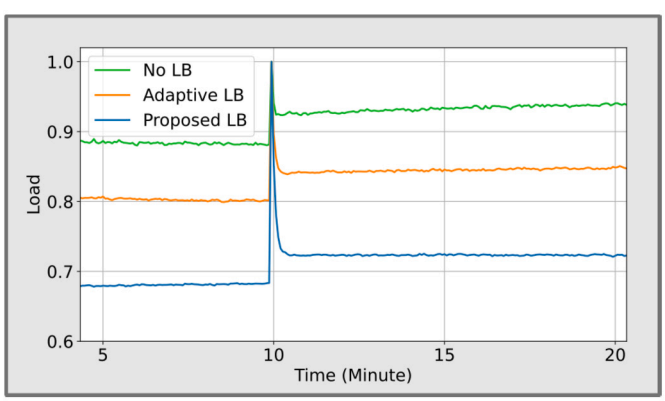

Fig. 9. Load change of the mBSs when the number of users is suddenly increased by 10% at time $t = 10$.

5. Conclusion

In this paper, an LB aware HO algorithm is proposed for B5G HetNets in order to improve the network performance in terms of load distribution. We model user mobility in terms of destination region attraction using a discrete-time Markov chain. In addition, a stochastic geometry-based HetNet topology is proposed to emulate a real BS deployment. Furthermore, we consider a more UE-dense region, such as a central location where people tend to gather more densely. The proposed algorithm is shown to improve network performance by avoiding heavily loaded BSs, and shifting the load from macro BSs to small BSs, as well as from the BSs within the attractive region to the other BSs. Therefore, the proposed algorithm can be considered as a promising solution to HO management in B5G networks with high mobility and dynamicity.

CRediT authorship contribution statement

Abdussamet Hatipoglu: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Mehmet Akif Yazici:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Mehmet Basaran:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Mine Ardanuc:** Methodology, Investigation. **Lutfiye Durak-Ata:** Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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